Ensembled U-Nets with Connectivity-Aware Post-Processing and Multi-Scale TTA

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

We segment solar filaments in GONG H-α images with a two-member, 5-fold-CV ensemble of U-Nets (EfficientNet-B4 and RegNetY-040 encoders) trained on a disk-registered, limb-corrected, soft-consensus target, followed by a learned connected-component post-processing pipeline (hysteresis thresholding, a shape-gated boundary-contrast scorer, and skeleton-aware mask regrowth) and 4-flip + 1.25×-upscale test-time augmentation. Our primary submission reaches a 5-fold day-grouped cross-validation Panoptic Quality (PQ) of 0.4297 ± 0.0048 (0.4298 pooled) – against a measured inter-annotator agreement ceiling of only 0.3399 on this dataset, and 0.37 on the public leaderboard. Every candidate change was tested against pre-registered statistical decision bars; the multi-scale TTA was the largest single gain, while more than a dozen others (a second model seed, a third encoder, a learned scorer v2, higher-resolution training and self-training on unlabelled GONG images among them) were rejected on the evidence. All 15 trained checkpoints are released with the code, and the submission can be regenerated from them. Code and weights: https://github.com/nicky5851/solar-k.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

The single most important fact about this task, established early and confirmed independently twice (Sec. 2.1; a transparent public baseline, discussed below), is that two expert human annotators agree with each other at only PQ = 0.3399 on this dataset. Every number in this report should be read against that ceiling, not against 1.0. We also found, by inspecting public Kaggle notebooks for this competition, that the public leaderboard's evaluator does not penalize missing or near-empty predictions correctly: an all-empty submission scores a public 0.88, and one RLE blob of precomputed answers (no training loop, exploiting a public copy of the underlying MAGFiLO dataset) scored "0.55+". A later audit of the public notebooks found none that genuinely scores above about 0.40 publicly. We therefore never optimized against the public leaderboard and treat any public score above about 0.45 as unverified; every decision in this report was made on local, day-grouped 5-fold cross-validation alone. We did not exploit the evaluator or use any labels beyond the competition training set.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Dataset

The training set has 707 grayscale JPEG images and the test set 180, all 2048×2048, uint8, single-channel, 702 MB on disk in total. A SHA-256 check found zero exact duplicates within the train set, within the test set, or between the two.

Annotations are provided as a COCO-style JSON file. Its 1154 `images` entries map to only 707 unique physical train files: one entry per annotator of the same image (411 images with one annotator, 145 with two, 151 with three). The file contains 8199 instance (polygon) annotations across three categories that are actually used – Right (2590), Unidentifiable (3074), and Left (2535) – with a fourth category, Ambiguous, defined but unused. Instances are small and elongated: median area 1230 px^2 (IQR 670–2440), median elongation (skeleton length / $\sqrt{\text{area}}$) 3.75, and a median minimum width of 2 px. Among a single annotator's own instances on the same image, 0.5% touch or overlap a neighbour.

Where an image has two or three annotators (296 images, 1196 directed pairwise comparisons), we computed an inter-annotator agreement reference using the competition's own PQ/IoU-matching rule: global PQ = 0.3399 (SQ = 0.6337, RQ = 0.5364; TP= 4292, FP= 3709, FN= 3709). Matched-pair IoU has mean 0.634 (std 0.077), and the disagreement is dominated by instance-identity mismatches rather than boundary quality: 253 one-to-many and 253 many-to-one instance correspondences, plus 2840 unmatched instances with no IoU-> 0.5 counterpart.

We split the 707 training images into 5 folds grouped by *observation day* (not by the JSON's `image_id`, so that different annotators of the same physical image always stay in the same fold), approximately balanced by image count and by observing site; the automated split check reported no violations.

2.2 Preprocessing

Each image is first registered to the solar disk: an Otsu threshold followed by the largest connected component gives a disk centre and radius (radius = $\sqrt{\text{area}/\pi}$; mean fitted

radius 905 px across the training set). We then correct limb darkening by dividing the image by its own per-image radial median-intensity profile, computed in 100 radial bins restricted to disk pixels and linearly interpolated across any sparsely populated bins; a numerical floor (10% of the profile's 5th percentile) prevents the ratio from blowing up near faint edge bins. The limb-corrected image is then percentile-clipped (1st/99th percentile, computed over disk pixels only) and rescaled to $[0, 1]$; pixels outside the fitted disk are set to 0. This deterministic pipeline is applied once per physical image, and its float32 output is cached to disk at float16 precision together with the fitted disk parameters, so repeated reads during tuning and training do not repeat the fit or the correction.

For semantic-segmentation training targets, each physical image's annotators are first rasterised into per-annotator binary masks (the union of that annotator's own instances); the training target is then the per-pixel *fraction* of annotators marking that pixel, so a single-annotator image yields a binary target and a multi-annotator image yields a soft target reflecting annotator (dis)agreement. Per-annotator instance masks are kept separately for evaluation rather than folded into this target.

2.3 Model

We train a standard U-Net [3] (via `segmentation_models_pytorch`) with a single input channel and a single sigmoid output channel, over two ImageNet-pretrained encoders chosen to preserve stride-2 (rather than stride-4) features from the first encoder stage – filaments are 2–4 px wide, and a coarser first stride destroys them before the decoder can recover them. The two encoders, EfficientNet-B4 and RegNetY-040, are each trained independently with 5-fold day-grouped cross-validation (10 models total: 2 encoders $\times$ 5 folds), on random 512×512 crops with augmentation, BCE + soft Dice loss against the soft-consensus target of Sec. 2.2, an EMA of the weights (decay 0.999, used for all inference), and checkpoint selection by validation PQ (computed via flip-TTA inference against instance-level ground truth, not pixel Dice) every 1000 of 6000 iterations.

At inference, each encoder's 5 fold-models are applied with 4-view flip TTA (identity, horizontal, vertical, both) plus a $1.25\times$ upscale-only augmentation (never downscale, since that destroys 2 px filaments), averaged into one probability map; the two encoders' probability maps are then averaged into a single ensemble map. A second same-architecture seed and a third (weaker) encoder were both tried as additional ensemble members and rejected (Sec. 3) – the two-member, two-encoder ensemble above is what is submitted.

2.4 Post-processing

The ensemble probability map is converted to instances in four steps: (1) hysteresis thresholding (a low and a high threshold) with a morphological closing, whose connected components become candidate instances; (2) an optional shape gate on component elongation and length; (3) a learned scorer (logistic regression plus a depth-3 gradient-boosted classifier on per-component shape and probability-contrast features) that estimates $P(\text{match}) \times E[\text{IoU} \mid \text{match}]$ and keeps components above a tuned cutoff; (4) a watershed *regrow* step that re-expands each kept component along the probability map down to a probability floor within a fixed dilation limit, recovering boundary pixels the hard threshold cut off. For every cross-validated number in Sec. 3, all of these hyperparameters are refit inside each fold on that fold's training images only, never on the held-out fold. For the submission, the chain is fit once on all 707 out-of-fold training predictions and frozen (hysteresis 0.40/0.99, closing radius 3, no shape gate, scorer cutoff 0.325, regrow floor 0.4 with 4 dilation steps). Figure 1 summarizes the full pipeline.

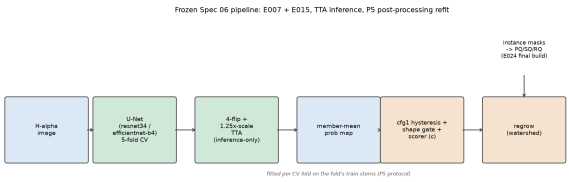


Figure 1: The frozen pipeline: two U-Net encoders, each 5-fold cross-validated, with flip+scale TTA at inference, averaged into one probability map, then a per-fold-refit hysteresis/scorer/regrow post-processing stack.

3 EVALUATION

All PQ/SQ/RQ numbers below are local, 5-fold day-grouped cross-validation (CV split by observation day, not by image, so all annotators of one image stay in one fold), with every post-processing hyperparameter refit per fold on that fold's own training images only (Sec. 2.4). Two of the ten fold-checkpoints (the RegNetY-040 encoder's folds 3 and 4) were lost after the original scoring run and retrained from scratch with the original configs and seed. Re-running inference from an existing checkpoint reproduced its stored probability maps bit-for-bit (322 files, maximum difference 0), so inference itself is deterministic; the retrained folds differ only through GPU training nondeterminism. The rebuilt pipeline's PQ differs from the originally scored run by -0.0004 pooled (at most 0.0038 on any fold), inside our pre-set ± 0.005 reproducibility bar, and the numbers below come from this rebuild. We never selected on, or report a final decision from, the public leaderboard (Sec. 1). Every candidate change below was accepted or rejected against a pre-registered statistical bar (minimum mean ΔPQ, fraction of folds positive, and a bootstrap 95% CI excluding 0 on the paired per-fold difference), not by inspection.

The full decision table, with each experiment's ΔPQ and the bar it missed, is in `reports/08_final_writeup.md` in our code repository. The changes adopted were a wider post-processing grid search, using RegNetY-040 rather than

Table 1: Progression of the submitted pipeline. PQ is pooled across all 707 train images' out-of-fold predictions; Kaggle LB is the public leaderboard (unverified, Sec. 1). †Our backup submission; rebuilt from the released weights it scores 0.4176.

Model	PQ	SQ	RQ	LB	Notes
Inter-annotator	0.3399	0.6337	0.5364	–	human ceiling
E001 (classical)	0.0346	0.6317	0.0547	–	dark-feature baseline
E002 (1 U-Net)	0.2691	–	–	–	single fold, no ensemble
E004×5 (1 enc.)	0.3948	0.676	0.584	0.35	5-fold CV, no 2nd encoder
+2nd encoder	0.4086	0.680	0.601	0.37	2-enc. ensemble†
Final (submitted)	**0.4298**	**0.681**	**0.631**	**0.37**	4-flip + 1.25× scale

ResNet-34 as the second encoder (keeping all three encoders was net-negative at every ensemble weight once the grid widened), and the TTA change in Table 1's last row. More than a dozen others were rejected, each on a pre-registered bar, including: a second same-architecture model seed (passed a cheap early gate but the full ensemble-promotion bar needed mean $\Delta\text{PQ} \geq 0.010$ on $\geq 4/5$ folds, and it only reached $+0.0023$ on $3/5$); a learned scorer v2 using cross-member agreement features, which won its own inner-split selection in $4/5$ folds but lost the held-out fold on every one (-0.0019 mean, $0/5$ folds positive) – a clean overfitting signature; a connectivity-trained encoder with a skeleton-recall loss term, which failed its own cheap gate before the full 5-fold training cost was paid; and a fragment-merging post-process step that fixed the error type it targeted (one-to-many/split errors fell) but increased many-to-one/merge errors, violating an explicit project constraint. Two late levers, training at 1.5× input resolution and self-training on pseudo-labelled unlabelled GONG images (images only, no external labels), both failed their early gates (ΔPQ -0.002 to -0.008). One experiment – adding a 1.25× upscale pass to an already-present flip-only TTA – was run as a controlled pair specifically to isolate which half of the change was doing the work: 4 flip views alone changed PQ by -0.0009 (noise); adding the scale moved it by $+0.0082$ pooled across all 5 folds (CI $[0.0042, 0.0127]$), the single largest and cleanest-signed gain in the project.

Distributions. Over 1154 annotator–image pairs, per-image pixel Dice has mean 0.654 (median 0.688, IQR 0.598–0.755) and per-image pixel IoU mean 0.503 (median 0.524, IQR 0.427–0.606). Pooled over all 707 images the final pipeline has TP = 4742, FP = 2080, FN = 3457. PQ varies by only 0.037 across the six GONG sites (0.417–0.454).

Error decomposition. 69.5% of all false negatives have no overlapping prediction at all (undetected filaments), concentrated in the smallest quartile of ground-truth instance area (PQ 0.31 in that quartile vs. 0.50 in the largest, driven by RQ 0.46 vs. 0.71 while SQ stays at 0.66–0.70). One-to-many (fragmentation: 219 FN, 450 FP) and many-to-one (merging: 209 FN, 102 FP) errors are a comparatively small share (669 and 311 FN+FP respectively, vs. 3347 for undetected filaments) and were essentially unmoved by the TTA change – the gain is a genuine recall improvement on faint/small filaments, not a reshuffling of which error type dominates. Figure 2 shows this directly: a typical image, the worst fragmentation case, the worst merge case, and a faint filament recovered by TTA that the backup ensemble missed entirely.

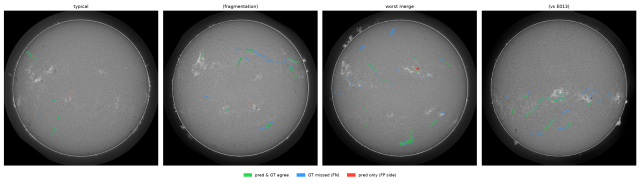


Figure 2: H-alpha overlay gallery, out-of-fold predictions only. Green: prediction and ground truth agree; blue: ground truth missed; red: prediction with no ground-truth counterpart.

Weaknesses and future work. The smallest-filament recall gap (PQ 0.31 in the smallest GT-area quartile) is the largest remaining lever and was only partly addressed this cycle; a scorer that actually generalizes past the inner-split overfitting we observed, or a loss term that more directly rewards small/faint-instance recall during training (rather than only at TTA/post-processing time), are the most promising untried directions.

Code and reproducibility. Code, configs, the 15 trained fp16 EMA checkpoints (as a GitHub release, sha256-checked) and both submission files are at https://github.com/nicky5851/solar-k (tag `v1.0-final`). Running `download_weights.py` then `predict.py` regenerates the submission: from the released weights it matches the submitted file's instance count on 172 of 176 test images, with mean IoU 0.9998 on those, and is bit-identical on 73.5% of images. The remaining differences come from GPU nondeterminism.

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